GISMA UNIVERSITY OF APPLIED SCIENCES

Project Management Essay

Student Accommodation App Development

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Declaration:

We confirm that this submission is our own original work, that every external source has been acknowledged using the Harvard referencing system, and that no part of this assignment has been submitted for any other assessment. We have read GISMA's policy on academic integrity and understand that plagiarism is treated as a serious offence.

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Introduction

Higher education is one of the largest drivers of international movement, and the moment a student decides to study abroad, the very first practical problem they face is where to live. Finding a safe, affordable place close to campus often takes longer than the visa or the course application itself. The aim of this paper is to set out the project management plan behind a mobile application that we intend to build for this market: a student accommodation app that connects verified landlords and university housing offices to students who are looking for short-term or long-term rooms (Silvius and Silvius, 2015).

Existing websites and apps cover parts of the problem, but each tends to fall short somewhere. Some focus only on private rentals, others list student dormitories but only in one city, and several are written in a single language that international users struggle with. A student arriving in a new country usually wants something very specific: a listing that matches their budget, sits close to the university, and can be trusted without any face-to-face meeting (Sharma and Coleman, 2026).

Our product is therefore positioned as a focused tool rather than a general rental marketplace. Trust, language clarity and search quality matter more than the size of the catalogue. As a finished application it should give the user a short, well-filtered list that they can act on quickly. Building a mobile app of that kind is not a small task: it needs a defined project structure, a controlled budget, a tested release cycle and a credible risk plan, and that structure is what this paper describes (Silvius and Silvius, 2015).

Project Overview

Developing the Student Accommodation App is more involved than writing code. The project covers requirements gathering, system design, development, quality assurance, deployment and post-launch support. Each of these stages is broken down and assigned to a named team member in the sections that follow. The overall goal is to deliver an effective application on time and within budget, which together is the project triangle that project managers are expected to keep in balance (Sharma and Coleman, 2026).

Student Accommodation App Development Project

The project is built around five members, each responsible for a clearly defined area. The product itself is an app that lets students search for apartments, shared houses and studio rooms in verified locations, where the landlord or the university housing office has already been screened. The scope of the application is deliberately narrow: it is intended to be quick to install, easy to search and dependable in its listings, rather than a comprehensive real-estate platform (Odunuga and Olayinka, 2023).

Project Structure

Responsibility within the team is distributed by function rather than by feature. One member acts as Project Manager, controlling the budget and coordinating the team. A second member leads interface and system design. A third member implements the application in Python and is responsible for fixing defects that arise during testing, which the whole team and selected stakeholders carry out before release.

A fourth member runs the support function, with a target turnaround of twenty-four hours and an additional remit covering risk handling such as fraudulent listings or suspicious user behaviour. The fifth member owns reviews, ratings and feedback, so that the team has continuous evidence of what is working and what is not. Beyond the team itself, the wider stakeholder group includes students, parents, landlords, property owners and the local government, all of whom benefit from a more transparent rental market (Sanyaolu et al., 2023).

Project Purpose

The purpose of the project is to give students a safer and faster route into local accommodation. Many students move country to study, and finding somewhere cheap, secure, close to campus and occupied by reasonable housemates is often the most stressful part of their first month.

Although governments, universities and non-profit organisations all run programmes that try to help international students, the support is usually fragmented across different websites and difficult to navigate from abroad. The platforms that do exist tend to overlook the specific needs of international students, which leaves obvious gaps in the rental process and exposes new arrivals to scams and unnecessary stress (Odunuga and Olayinka, 2023).

Project Scope

In scope. The app will list verified apartments and houses with photos, full contact details, an address, a map, the rental cost, registration information and any utilities included. Every listing is checked by our team before it goes live. We will also include review and rating sections for each property, so that students can sort accommodation by reliability, distance from campus or price.

Listing data will be sourced through partnerships with landlords, university housing offices and selected accommodation providers. Our reference point during the design phase is the Studierendenwerk Stuttgart app, which already offers a useful range of student services such as canteen menus, laundry booking and one-to-one consultations. The Stuttgart app shows what is possible when dormitory services are bundled into a single mobile interface; its main weakness is that the interface is dated and the entire application is in German, which limits its usefulness for international students (Bhandari, 2024). Our app addresses both of these gaps directly, with a multilingual interface and a modern UX layer.

Out of scope. The application will not handle in-app payments. Rent transfers and deposits stay between the student and the landlord, which removes one of the main vectors for fraud. The rental contract itself is also outside the scope of the app; both parties remain responsible for the legal agreement, with the app acting only as the discovery layer.

Project Deadline and Schedule

The schedule sits at the centre of the project. The build runs from 1 June 2026 to 18 June 2026 in the planning version of the Gantt chart, and a longer sixteen-week version is used for the full development cycle. A documented timeline gives the team a shared view of what has to be ready by when, allows resources to be allocated more efficiently, and forces every member to remain accountable to the same set of milestones. The first phase of the cycle is requirements gathering: short interviews and survey responses from students and property owners that help us define exactly what the app must do.

The second phase translates those requirements into system design and a user-interface specification. The third phase covers coding, internal testing and defect resolution. Deployment to the Apple App Store and Google Play follows after successful testing, with post-launch monitoring extending into the support phase. Milestones are reviewed weekly to keep the project on track, because fixed-deadline projects fail more often than not when the team allows variances to accumulate quietly between checkpoints (Meléndez and Anyosa, 2008).

Project Plan

The project plan is the working document that explains how the application will be built, controlled, executed and tracked. It captures the scope, the deliverables and the assignment of responsibilities, so that every team member knows what is expected of them at any point. The project manager is the single point of contact for stakeholders, and risk management is embedded into the plan from the start: each technical area is given a backup approach so that, if a component fails during integration, work can continue in parallel rather than stop entirely (Kerzner, 2017).

Activity Groups and Networks

Activity groups and network diagrams are used to keep the schedule visible. The app follows the standard project lifecycle - initiation, planning, execution, monitoring and closure - and within each phase a network diagram shows which activities depend on which others. The design phase cannot begin until end-user requirements have been collected; coding cannot begin until stakeholders have approved the design; testing cannot begin until the application has been built (Maylor, 2010).

A project network is typically drawn as a flowchart of boxes and arrows, where each box is an activity and each arrow shows the dependency between two activities. If any activity in the chain is delayed, the network makes it immediately clear which downstream tasks are also affected. Grouping similar tasks into a single activity category, rather than tracking dozens of micro-tasks individually, keeps the diagram readable without losing precision.

Project Budget

A workable budget is essential for the app to reach launch. The team has to know, at every phase, how much has been spent and how much is still available. The largest cost lines are system design, software development, testing, cloud hosting and project management, with smaller lines for licences, security audits and marketing. An emergency reserve sits on top of the planned spend, because every software project of this size encounters at least one unplanned cost that the original estimates did not predict (Baccarini, 2018).

Tracking is as important as estimation. Without weekly reviews of actual versus planned spend, even a well-built budget can drift until the project runs out of cash before launch. The team uses a shared spreadsheet linked to the schedule, so that whenever an activity is marked complete the project manager can see both the time variance and the cost variance for the same task. This is the operational backbone of Earned Value Management, applied at a small-team scale rather than the enterprise scale described in PMI (2021).

Cost Classification

Project costs are sorted into four categories so that the team knows exactly where every euro is going. Direct costs cover the core build - development hours, test environments, design tools - and are tied to a specific deliverable. Indirect costs cover administrative effort, internal communication and general team overhead, none of which produces a single visible deliverable but all of which are still necessary for the project to function.

Fixed costs, such as software licences and the domain registration fee, are the same regardless of how long the project runs. Variable costs, such as cloud-hosting fees and pay-as-you-go services, scale with usage and therefore need closer monitoring as the user base grows. The emergency reserve sits separately, is not drawn down for predictable lines, and is released only against a documented risk event (Kádárová, 2013).

Full Business Case

Finding accommodation as a student is stressful and time-consuming, and the friction in the current market is bad for everyone involved. Many students rely on closed Facebook groups or informal agents whose listings are inconsistent and occasionally fraudulent. Landlords, in turn, struggle to find reliable tenants without paying agency fees. The business case for our app is that a single trusted platform - with verified listings on one side and verified tenants on the other - reduces friction for both groups and produces a healthier rental market close to each campus (Sharma and Coleman, 2026).

Statement of Work (SOW)

The Statement of Work captures the terms under which the app is built. It is the formal agreement between the team and the stakeholders, and it includes the scope of the application, the timeline, the deliverables, the responsible team members and the reporting cycle. The SOW is signed off at the start of the project and is updated only through a controlled change-request process (Cole and Martin, 2012).

Within the SOW, the development team commits to design, build, test and launch a fully functional application on both iOS and Android. The scope explicitly covers user-interface design, user-experience flow, back-end development, the application programming interfaces that move data between services, and a screening process that filters out misleading listings before they are published.

Project Deliverables

Deliverables are the measurable outcomes of the project. They include the published iOS and Android applications, the UI/UX prototypes used during the design phase, technical documentation for the back end, an administrator handbook for partner universities, and a short user training guide aimed at first-time students. Every deliverable is reviewed by the project manager, and signed off by the relevant stakeholder, before it counts as complete (Butler, Vijayasarathy and Roberts, 2020).

WBS and OBS

The Work Breakdown Structure (WBS) divides the project into successively smaller pieces of work, until each leaf node is small enough to be estimated and assigned. The

Organisational Breakdown Structure (OBS) then maps each piece of work to a person or a function, so accountability is never ambiguous. The project manager allocates resources and communicates directly with stakeholders, while the design, development and QA teams report into their respective leads. Daily collaboration is encouraged across the lines of the OBS to keep the build moving (Schwalbe, 2019).

Stakeholders and Benefits

The stakeholder map is wider than it first appears. The primary internal stakeholders are the project sponsor, the project manager and the development team: the mobile developers who write the iOS and cross-platform code, the UI/UX designers who shape the visual language and the screen flows, and the QA engineers who act as the final checkpoint before each release. Secondary stakeholders include university administrators, private landlords and, most importantly, the students who will use the app every day. Each group has different success criteria and the project plan accounts for all of them (Sanyaolu et al., 2023).

The benefit of the application is structural. Students gain a single, verified place to search for accommodation rather than spending weeks across multiple platforms. Landlords reach appropriate tenants directly, without paying an intermediary. The verification process means that the photographs, the addresses and the contact details a student sees on the app reflect a real property; students themselves go through a comparable verification before they can contact a landlord. The overall result is a cleaner two-sided market with fewer fraudulent listings and a shorter time-to-rent.

Project Community

A successful launch needs more than functioning code. The team produces a Product Requirements Document and an internal design blueprint before development begins. The application launches in both major app stores and ships with automated property verification and real-time chat between landlord and student. Documentation includes administrator guides for partner universities, security audit reports for future regulators, and a monthly community feedback session that captures what is working and what needs to change in the next sprint.

Risk Management

Risk management runs alongside every other phase of the project. Building a mobile platform that combines real-time listings, user authentication and a verified review system creates several categories of risk, and a structured approach is needed if the application is to be delivered on schedule, within budget and at the expected quality level (PMI, 2021).

Identifying Risk and Probability

Technical risks tend to be underestimated during the planning phase of software projects, which is why they so often cause expensive rework later in the cycle (Kerzner, 2017). For a system of this size, the probability of hitting at least one significant technical issue during development is rated high, somewhere around seventy per cent.

Financial risks come mainly from budget overruns and from new costs that appear when the scope changes or when a planned resource becomes unavailable. With a fixed budget, an unplanned expense can compromise the quality of the final product, and the probability of some budget deviation is estimated at around fifty per cent, especially if the timeline stretches past the original delivery date (Maylor, 2010).

Operational risks cover the team itself: a missing member, a skill gap, or poor communication between students who are also juggling other modules. Because every team member is balancing academic deadlines, the likelihood of small delays through workload conflict is treated as medium to high. External risks include regulatory changes around data protection - GDPR in Germany in particular - and movements in the wider market of accommodation platforms (Schwalbe, 2019).

Mitigation Pathway

Each identified risk is paired with a specific mitigation. Technical risk is addressed by adopting an agile build cycle, which puts working software in front of the team every sprint and exposes defects early. Code reviews are mandatory before merging, and the back-end database is encrypted at rest so that user data is protected even if a credential is leaked. Kerzner (2017) makes the point that proactive response planning materially lowers the cost of a risk event when it eventually happens, and that argument is the basis for keeping a ten-per-cent contingency reserve against the overall budget.

The schedule itself includes deliberate buffer periods between milestones so that small delays are absorbed without slipping the final delivery date. Scope is fixed early and any change must go through a written request, which protects the project against the kind of scope creep that PMI (2021) identifies as one of the most common causes of budget overruns. Operational risks are managed through clear role definitions, a weekly progress meeting and a shared project-management tool that shows the status of every task. For external risks, the team consulted GDPR guidance at the design stage so that data collection and storage are compliant from day one - far cheaper than redesigning after launch (Schwalbe, 2019).

Evaluation and Tracking

Evaluation and tracking are the processes that keep the project aligned with its original objectives through every phase. If a team stops monitoring its progress, deviations from the plan accumulate quietly and only become visible once they are too expensive to correct (Maylor, 2010).

Closeout and System Success

The evaluation framework for the Student Accommodation App rests on three measurable dimensions: time, cost and quality. Progress is checked against the Gantt chart every week and any variance from the baseline schedule is flagged immediately for team review. Earned Value Management techniques are applied alongside, so the team can compare the value of work actually completed against the budget consumed up to that point. This gives an objective view of whether the project is genuinely on track or only appears to be on track (Kerzner, 2017).

The closeout phase begins once the final release passes its acceptance criteria. It includes a formal review of every deliverable against the original scope, a sign-off meeting with stakeholders, archival of all project outputs and a lessons learned session in which the team records what worked and what should be done differently next time. PMI (2021) argues that a structured closeout is not only useful for confirming a project's outcome; it is the single most reliable way of carrying knowledge into the next project.

System success for the application is defined in two layers. Technically, the app must meet the functional requirements listed in the project scope and operate without critical errors after launch. From the user's perspective, success means a post-launch satisfaction score of at least eighty per cent in the first three monthly surveys. These two layers are tracked separately so that a technically clean release with poor adoption is not mistaken for a successful project, and so that high engagement on a buggy build is recognised as fragile rather than safe (Schwalbe, 2019).

Summary

The Student Accommodation App is designed to reduce the friction between two groups that already want to find each other: students who need a verified, affordable room and landlords or housing offices that need reliable tenants. The platform replaces the scattered mix of Facebook groups, untrusted listing sites and informal agents with a single mobile application that students can install before they leave home and continue to use after they arrive.

From a project management standpoint, the application is treated as a normal software project rather than a one-off student exercise. The team is small, the scope is deliberately narrow, the schedule is split into well-defined phases, the budget is categorised so that every line can be tracked, and the risk plan anticipates the technical, financial, operational and regulatory issues that a project of this kind is likely to face. Each section of this paper has documented the controls that apply to one of those areas.

The combination of structured planning, weekly tracking and a disciplined closeout phase gives the team the best chance of delivering an app that is technically sound and genuinely useful to the students it is meant to serve. The remaining work - the real test of the plan - is the build itself.

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